

Soldering 101

ARRL-American Radio Relay
League

Soldering

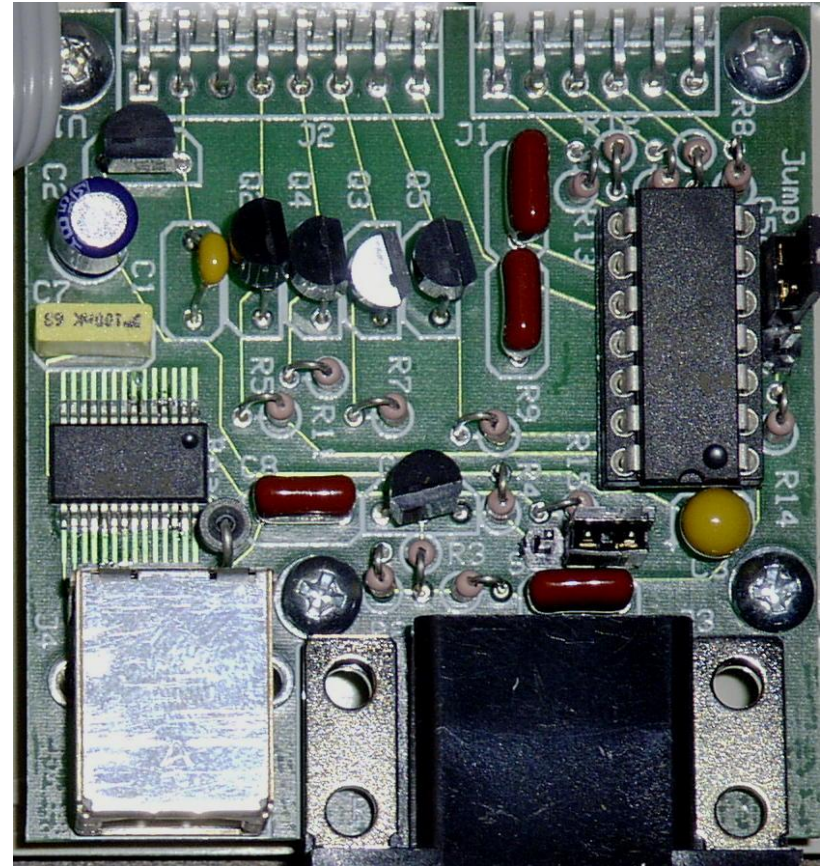
- The process of making an electrical connection by melting low-temperature metal alloys around component leads
- Soldering is just as much an “Art” as it is a “Science”
- We will:
 - Go over some soldering vocabulary
 - Illustrate graphically soldering techniques
 - Solder a few components on the TV Remote board
 - Complete the board as homework

Soldering Process in Brief

1. Turn on soldering iron and let it get to temperature
2. Clean soldering tip with a damp cloth or sponge
3. Install the component and hold in place with a physical connection
4. Heat the component lead and the adjacent circuit board pad – apply the right amount of heat (art)
5. Touch solder to the component lead and pad - melt the right amount of solder (art)
6. Remove solder
7. Remove heat (art)
8. Hold steady until cool
9. Inspect the connection (art and good eye)

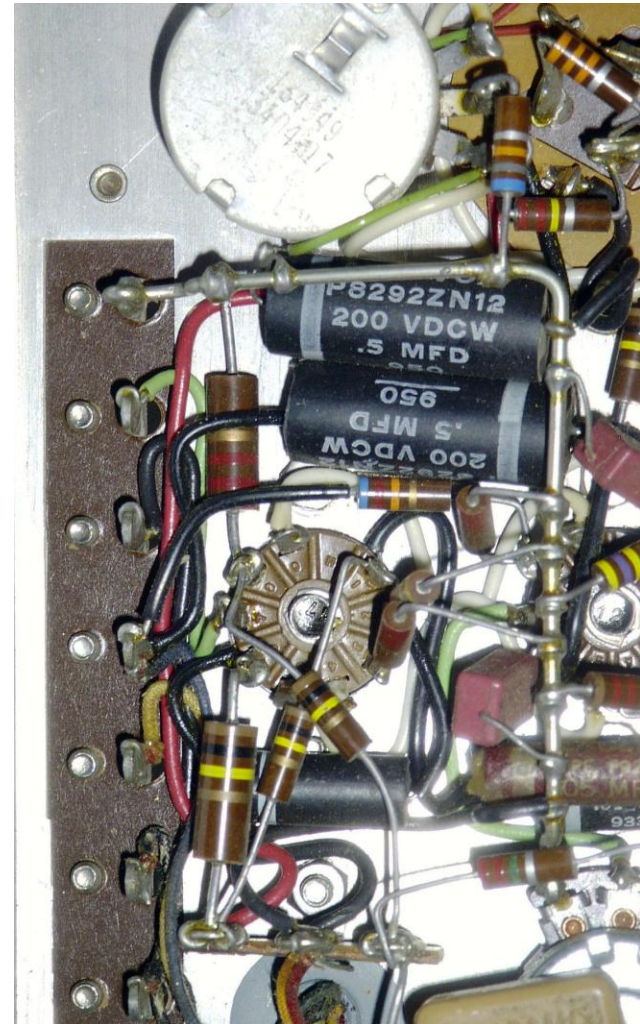
Vocabulary

- Circuit Board
- PCB
- Components
- Component side
- IC socket
- Pins



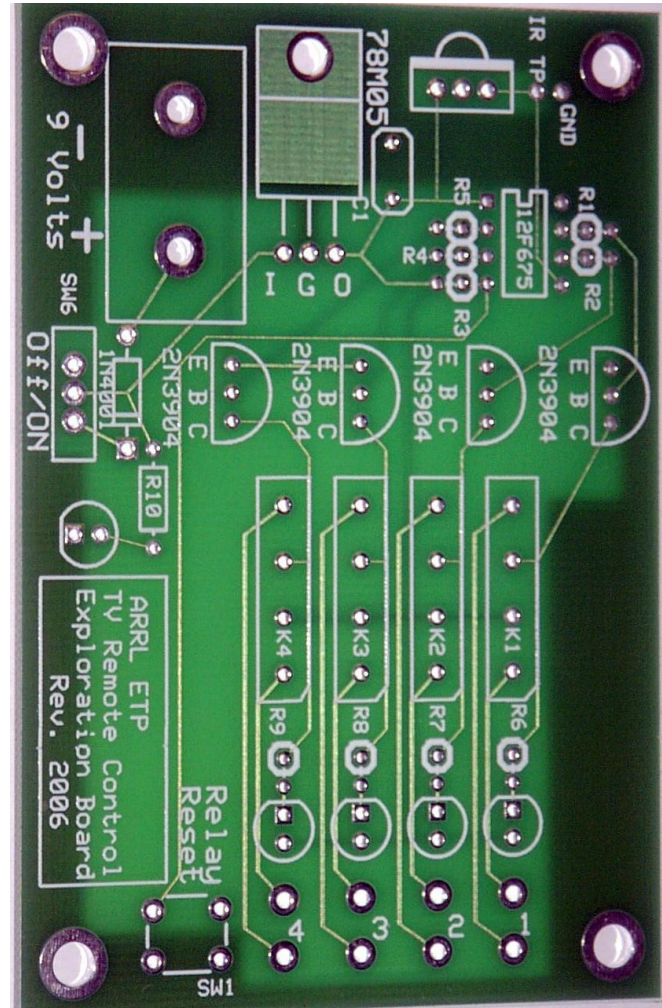
Vocabulary

- Point-to-Point wiring
- Physical (mechanical) connections
- Lugs
- Component leads
- Junctions



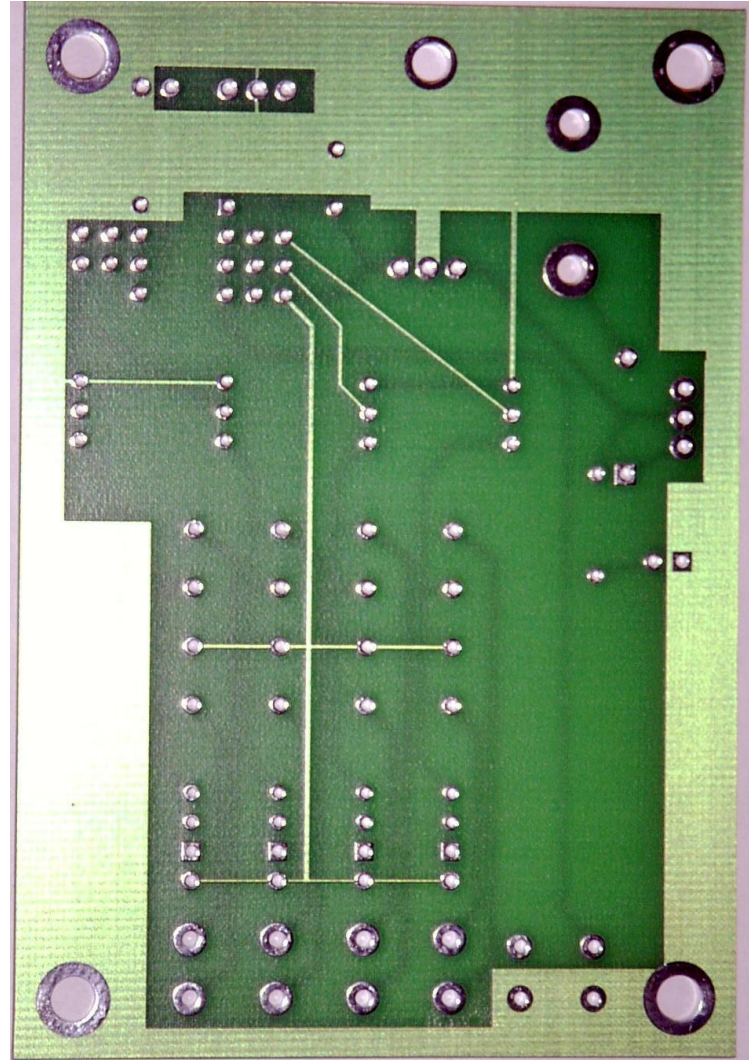
Vocabulary

- Component side
- Component outline
- Silk screen
- Double sided circuit board (top side)
- Top trace
- Solder mask



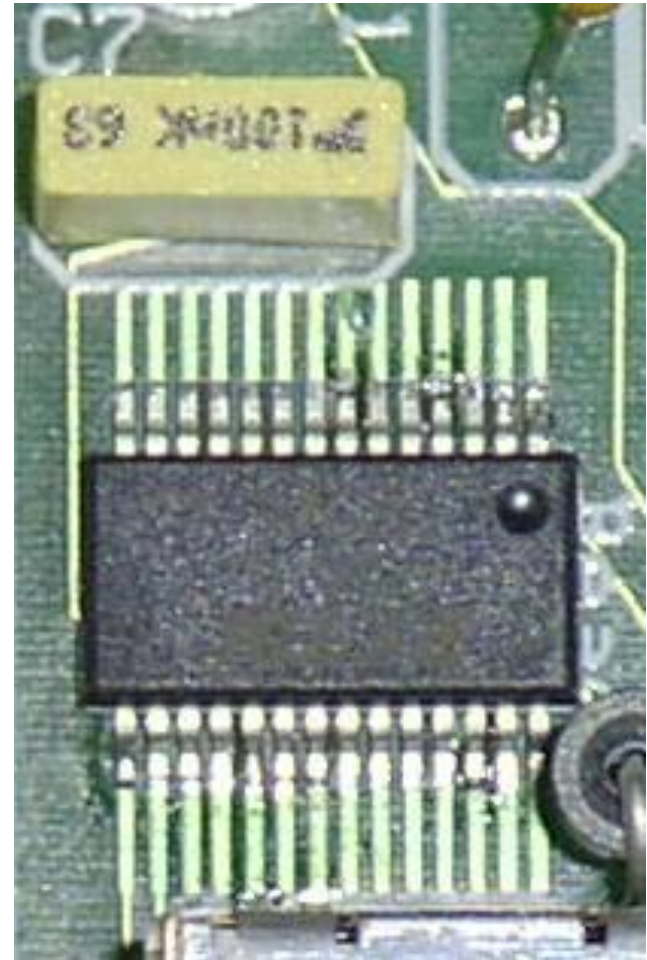
Vocabulary

- Double sided circuit board bottom side
- Solder pads
- Through hole
- Foil trace
- Pad
- Ground plane
- Soldering side
- Multi-layer board



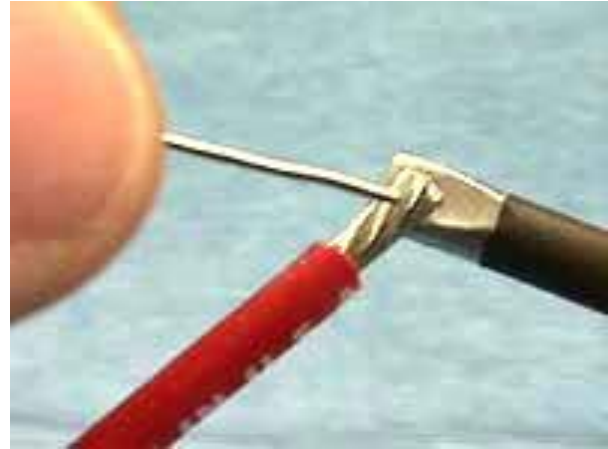
Vocabulary

- Surface mount component
- Pitch
- Pin #1 designation on IC



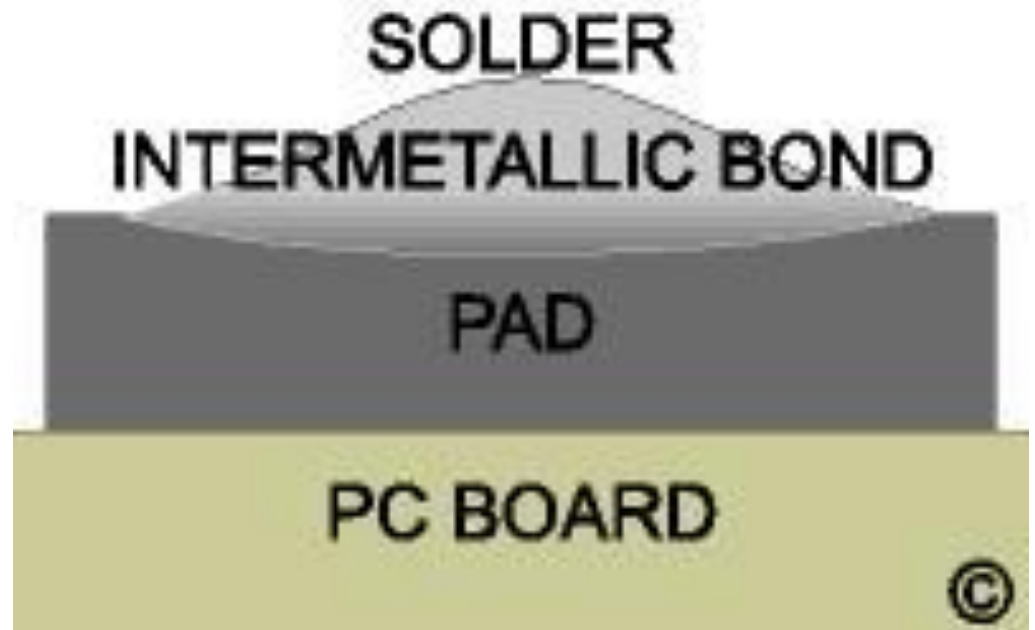
Vocabulary

- Tinning
- Wetting
- Surface tension
- Flow



Vocabulary

- Connection
- Volcano
- Shine or sheen
- Dull or crusty
- Cold solder joint
- Tack-solder



Vocabulary

- Soldering Iron
- Tip
- Soldering Station
- Tip cleaning surface

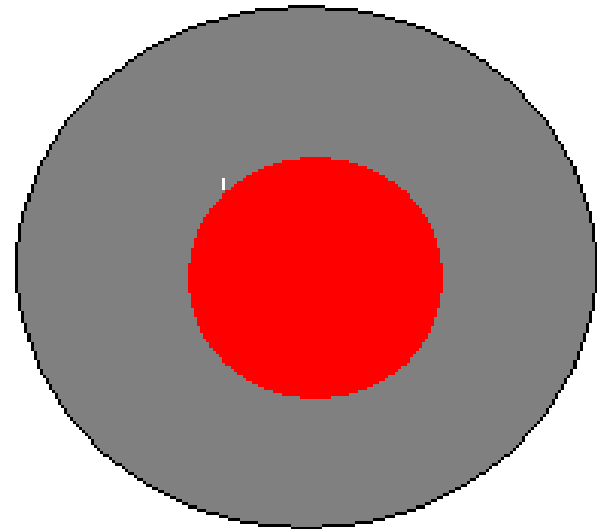


Solder Types

- Rosin core and Flux
 - Flux type is used for household soldering
 - Rosin core is used for electrical wiring (reduced corrosion potential)
- Rosin and flux are chemical compounds that clean the metal surfaces that are being soldered
 - The rosin or flux melts first to contact the joint
 - The rosin or flux float the impurities away from the metal surfaces and to the top surface of the solder joint
 - Produces the smoke you will see associated with soldering
- You will be using rosin core solder

Rosin Core Solder

- Outer tube of low melting alloy
 - Tin
 - Lead
 - (Industry is moving toward lead-free)
 - Silver
 - Copper
 - Other specialized metals
- Inner core is the rosin



Solder Specifications by % Alloy

Rosin Core Solder Rolls and Pocket Packs



73576



32117



170457



141778

* Organic core

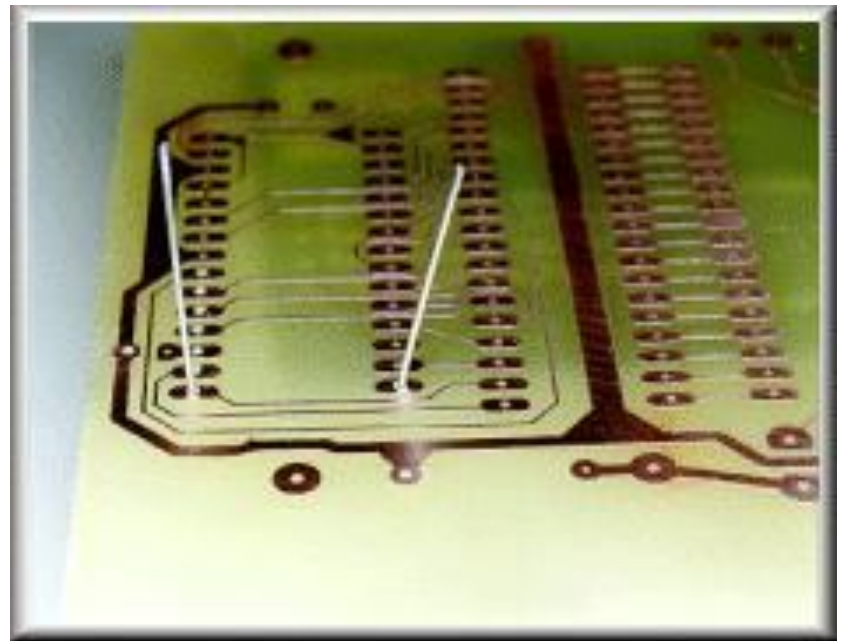
Part No.	Manufacturer	% Composition				Dia. (in.)	Wt. (lbs.)	Pricing			
		Tin	Lead	Silver	Copper			1	10	50	100
668078	—	99.3	—	—	0.7	0.04	0.6	\$10.45	\$8.64	\$8.20	\$7.79
668262	—	96.3	—	3.2	0.5	0.04	0.6	17.69	14.15	13.58	13.00
668271	—	96.5	—	3.5	—	0.04	0.6	15.95	14.40	14.20	13.97
668289	—	99.3	—	—	0.7	0.04	1.1	19.39	15.69	14.91	14.16
668297	—	96.3	—	3.2	0.5	0.04	1.1	29.95	27.50	26.75	25.99
668318	—	96.5	—	3.5	—	0.04	1.1	27.79	25.60	25.08	24.56
668326	—	99.3	—	—	0.7	0.04	2.2	38.75	31.39	29.82	28.33
668334	—	96.3	—	3.2	0.5	0.04	2.2	57.19	46.66	44.33	42.78
668342	—	96.5	—	3.5	—	0.04	2.2	65.45	62.18	59.07	56.12
141794	Super	60	40	0	—	0.031	0.55	5.95	5.39	4.85	4.35
141786	Super	60	40	0	—	0.031	1.10	7.95	6.95	6.25	5.82
141778	Super	60	40	0	—	0.031	2.20	14.69	13.95	13.33	12.80
170457	Golden	60	40	0	—	0.031	0.06	1.39	1.25	1.05	.99
73576	Kester (pak)	60	40	0	—	0.031	0.04	2.59	2.33	2.10	1.75
32117	Kester	60	40	0	—	0.031	1.00	13.55	12.25	11.05	9.95
73605	Kester (pak)	62	36	2	—	0.020	0.02	1.25	.99	.97	.96
112344	Kester	63	37	0	—	0.020	1.00	19.49	18.50	18.27	18.14
151474	Kester *	63	37	0	—	0.031	1.00	18.39	16.55	14.89	13.39

Soldering Process in Brief (again)

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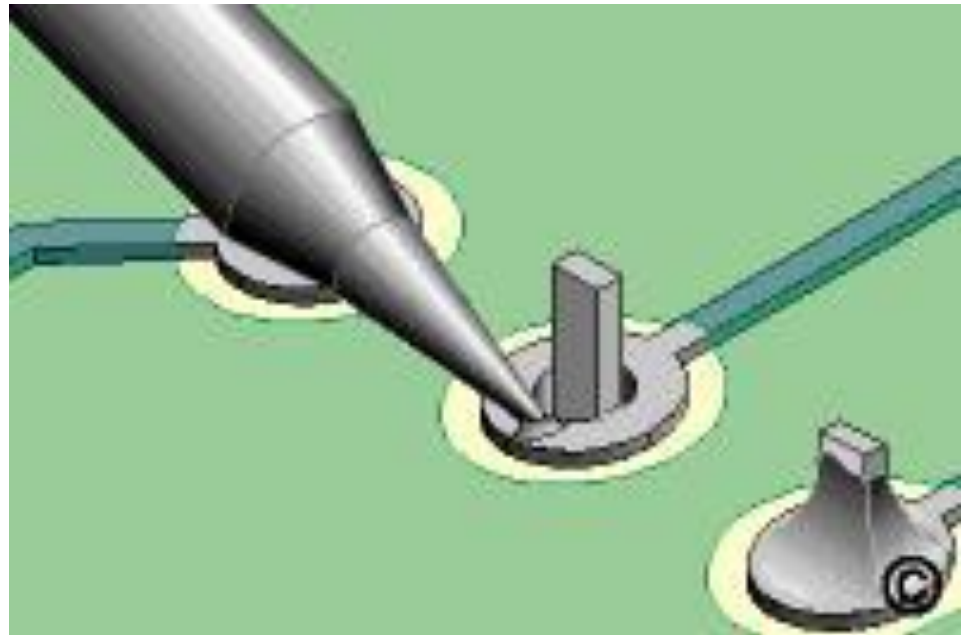
Soldering Process

- Insert component from the “component side” of the PCB and bend leads out to physically hold the component steady and in place
- When you can't bend leads – “Tack Solder” one lead to hold component in place

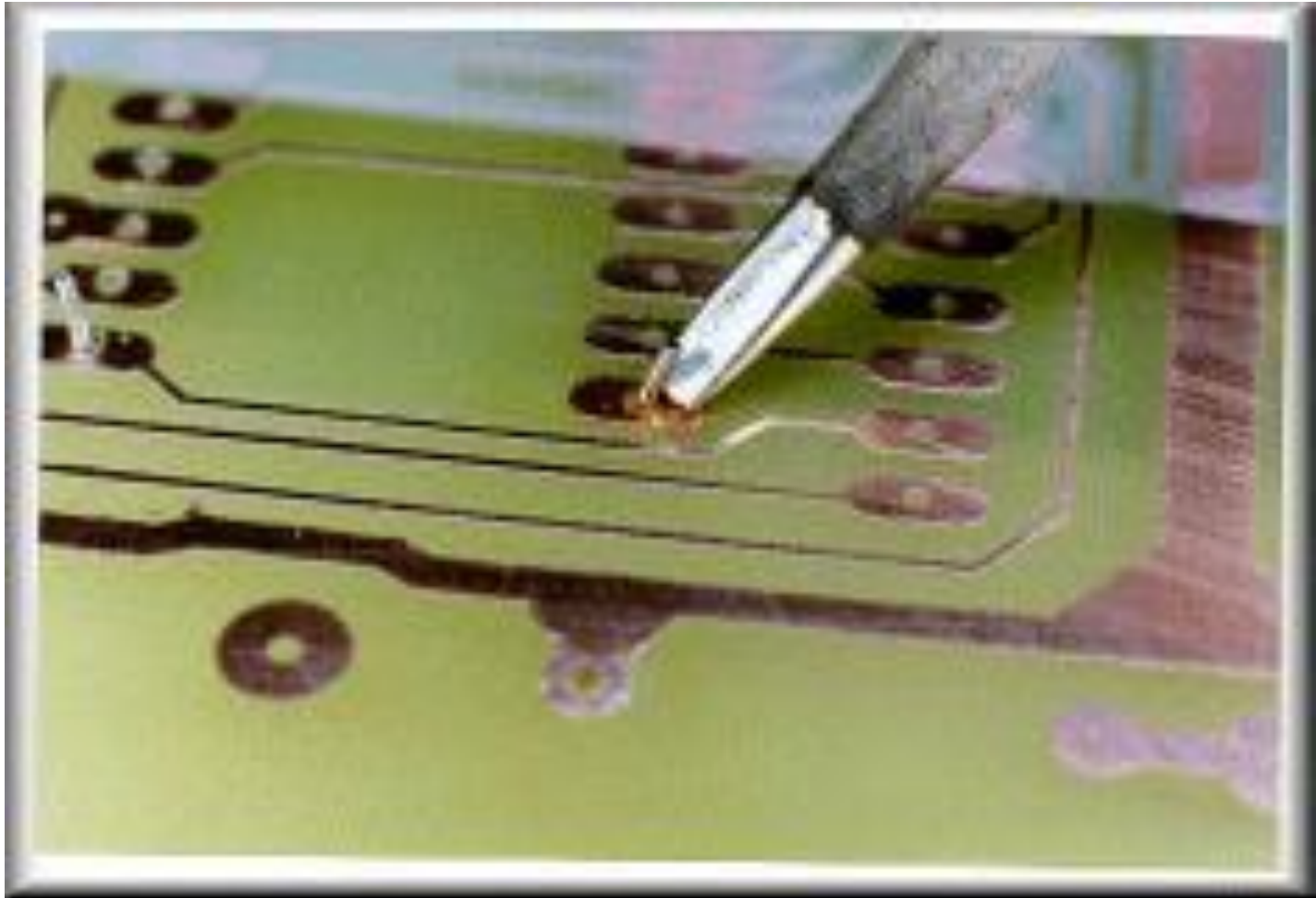


Soldering Process

- Touch the tip to the component lead **AND** the pad
- Apply just enough heat
 - TLAR (**T**hat **L**ooks **A**bout **R**ight)



Soldering Process



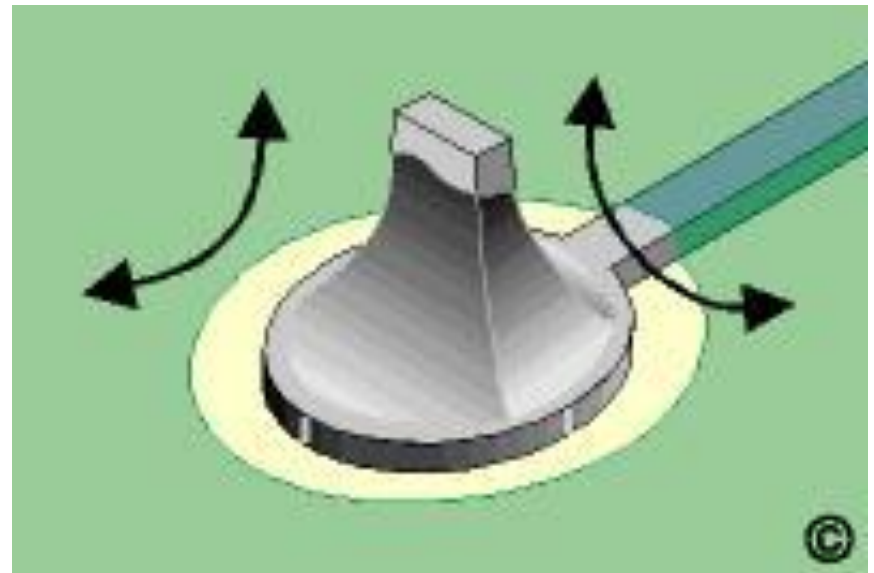
Soldering Process

- When you have headed the joint just the right amount (TLAR) touch solder to lead and pad (not to the tip)
 - Do as I say, not as I do!
 - Why I sometimes will use the tip?



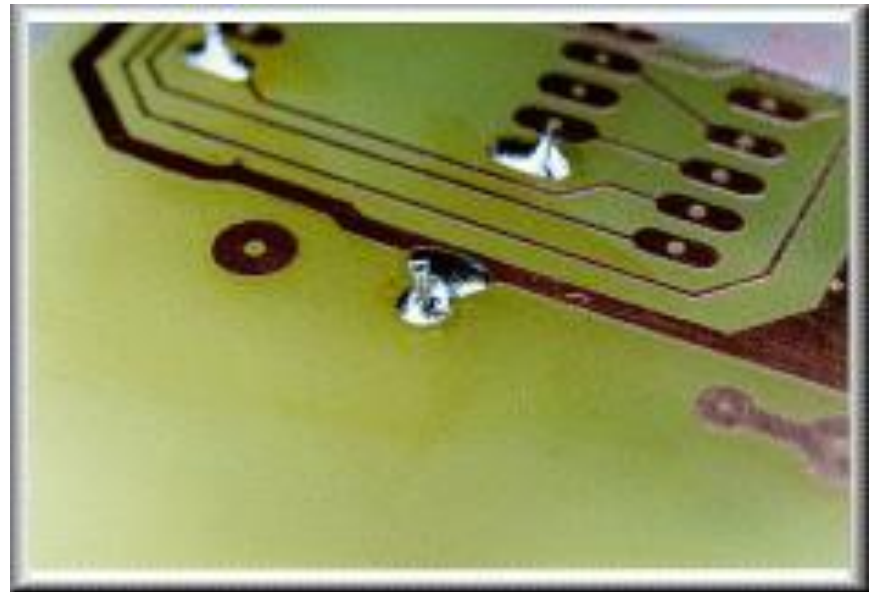
Soldering Process

- The melting solder will flow around the joint
 - Called wetting
 - Surface tension will produce a nice “Volcano” looking joint
- Keep the tip in place until TLAR



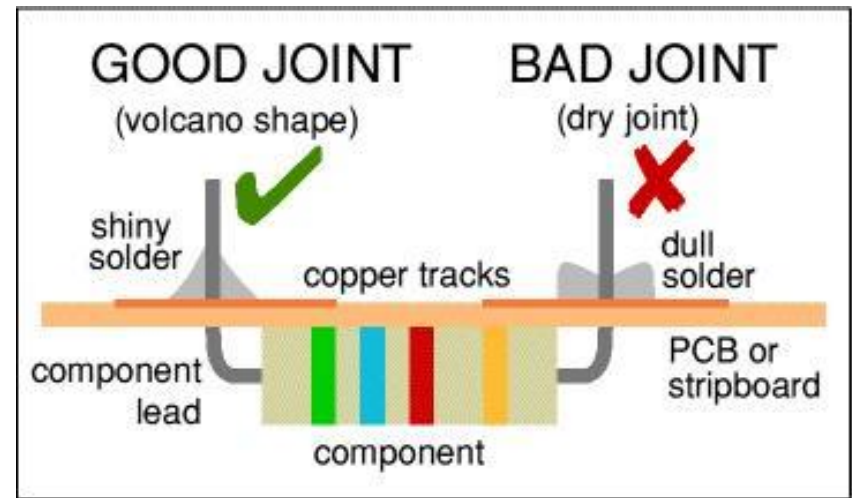
Soldering Process

- Remove solder then tip and allow joint to cool
 - Hold steady while cooling
- Inspect for good joint (TLAR)
 - Reheat “cold solder” joints
- Clip and remove excess leads (safety first, watch for flying leads)



Good vs. Bad Solder Joints

- Cold solder joints are probably the cause of most circuit problems!
- You need to use caution not to overheat a joint (TLAR)
 - But don't get overly concerned except for the most delicate components



Some Safety Concerns

- Don't solder while circuit is powered
- Use well ventilated and lighted work space
- Don't touch the solder tip – it's hot
- Static discharge protection – not a hazard for you but can trash some sensitive components
- Watch for flying leads when clipping excess

Your Turn

- Solder the LM358 Amplifier Board